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INTEGRATION OF A SIMULATION PLATFORM INTO A BUSINESS PROCESS MANAGEMENT TOOL

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Abstract

Information technology (IT) has already gone beyond being a supportive tool for businesses. Strong competition and increased costs force businesses act more efficiently and rapidly, so almost in every business, the operations are carried out in IT platforms where the processes are automated. These platforms are referred to as Business Process Management (BPM) Suites. With this motivation, most companies analyze and reengineer their workflows. At this point, it is vital to be able to monitor current or reengineered business processes with key performance indicators and keep the process flows under control. This introduces a challenge for the management to evaluate how any particular modification to any of the process workflows affects the performance measures. Therefore simulation becomes an essence in the contemporary management of businesses. In practice, major business process management softwares include some kind of simulation environment either as an embedded engine within the modeling component or through third-party integration. In this study a general purpose simulation platform is analyzed and a mapping algorithm is developed to integrate modeling capabilities of a BPM tool with this simulation environment.

Keywords: Business Process Management, Simulation

1 INTRODUCTION

In rapidly evolving business and technology era, Business Process Management (BPM) has emerged to handle ever-changing interactions between different parties and applications. In leading organizations, the critical importance of adapting new ways of doing business has been well understood by managers, but the challenge remains for them to justify and accelerate this transformation. Static modeling tools like MS Visio are widely used in modeling and studying business processes, but exclusion of stochastic behavior of entities within a process hinders the ability of process owners or analysts to predict how performance indicators and overall performance are affected. Once transformed into dynamic models, static flowcharts and workflow diagrams serve as much better decision support tools.

Simulation obviously plays an important role in modeling and (re)designing any business process. Simulation mainly allows users to determine how their process designs will work and provides a platform to assess risks before designing engagements. It helps to identify redundant steps, bottlenecks and opportunities that are involved in complex business processes. Furthermore, process visualization and animation are provided, allowing multidisciplinary team members to understand the model and communicate about it. By the recent improvements in modeling user interfaces, simulation may now be used by those who have little or no simulation background. On the other side, the use of

simulation in designing or reengineering a business process is well understood and the organizations need to identify cost minimizing and value added activities in designing their processes. These reasons have all been recognized by companies through struggles in real-world applications and BPM software developers started to incorporate simulation modeling features to meet these requirements. In a research conducted by Hall and Harmon (2005) as cited by Hlupic et al. (2007), most modeling tools offer discrete event simulation capabilities either as a part of the tool supported with a simulation engine or as a separate add-on module developed by themselves or through third party software development companies.

The simulation capabilities of the BPM tools where the simulation engine is a part of the tool are generally very limited. This is because testing a model dynamically by using different scenarios with well-defined and approximated input distributions or even with historical data is not the main focus of the vendors of these tools. Vendors defend this argument by stating that the workflows generated by the BPM tool are usually standard networks and many capabilities of the third party simulation software that provide detailed modeling and analysis are not often required. However it should be noted that the limitations in simulation capabilities might result in invalid models that yield misleading outcomes.

The second option in incorporating simulation capabilities to a BPM tool is to integrate it with a separate simulation software. This option is more costly and requires the knowledge on both platforms. Besides the extra coding effort to develop the necessary interfaces might be quite significant. Yet one should note the tradeoff between the increased costs and the broader functionality achieved by integrating a third party simulation software.

In this study, the ways of integrating a selected BPM tool and simulation tool are explored with the aim of observing the benefits and shortcomings of using a third party simulation software. The organization of the study is as follows: In the next section, recent literature is presented on the use of simulation in BPM tools. In Section 3, the selected BPM tool, Netflow and the simulation tool Arena are introduced by comparing their modeling and simulation capabilities. In Section 4, a mapping algorithm is developed to simulate the flows generated by the BPM tool in the simulation environment of Arena. In Section 5, conclusion and future work are stated.

2 LITERATURE SURVEY

In academic literature there are various definitions of BPM, all of which converge to the same point of improving the way of doing business and managing organizations (DeToro and McGabe, 1997; Elzinga et al., 1995). Workflow Management Coalition defines BPM as the practice of developing, running, evaluating and simulating business processes to affect the continued improvement of those processes. According to Association of BPM Professionals, BPM is a systematic approach to identify, design, execute, document, monitor, control and measure both automated and non-automated business processes to achieve targeted results consistent with an organization's strategic goals.

There are several studies that highlight the use of simulation for BPM. Serrano and Hengst (2005) develop a simulation model where they suggest performance metrics that can be used to depict business processes and information systems integration. Similarly, Barber et. al. (2003) study BPM and simulation from the perspective of manufacturing management. They state that except for a few small-scale processes, business process simulation implementations in manufacturing have had limited success and suggest a practical way forward until hardware and software limitations are overcome. Adamides and Karacapilidis (2006) present a framework for BPM that integrates a problem-structuring methodology, a knowledge management system and a modeling/simulation environment.

BPM systems bring along executable models; however in an early study, Paul et. al. (1999) argue that simulation will not become a mainstream modeling tool, mainly due to the lack of interface capabilities between simulation softwares and other modeling tools. Second reason is the requirement of a certain amount of expertise for building models with most of the simulation languages. However,

this problem is less significant in this new decade, since there are several simulation packages that offer modeling capabilities without the use of programming skills.

The lack of interface issue has been recognized by both the academia and the industry, leading to studies on integration between modeling tools and simulation. Harrell and Field (1996) present an integration study between Design/IDEF, a process mapping tool and ProModel, a simulation tool. Through modifying the underlying code of the process mapping tool, they manage to add entity attributes, input buffers; develop ways to model different entities; and generate additional data fields to capture dynamic information. Similarly, Srinivasan and Jayaraman (1997) come up with a model generator developed in SIMAN simulation language, called EMF-SIMAN. EMF stands for Enterprise Modeling Framework and consists of a methodology that distinguishes between the important aspects of an organization such as function, information and dynamics.

In a recent work, Waller et al. (2006) present a Java based simulation engine, L-SIM which is used to simulate processes constructed with BPM notation (BPMN). This commercial product provides specific objects to represent each BPMN shape and has an option of adding new objects to support simulation data analysis. This engine may be used with any BPMN compliant modeling environment via a BPMN compliant application programming interface (API). It is currently implemented in several BPM tools such as IDS Scheer, Telelogic and PNMSOFT (Lanner Group White Paper, 2009). The BPM tool considered in this study does not support BPMN; a tool specific integration platform is generated to observe the benefits and shortcomings of using a third party simulation tool.

3 SIMULATION ENVIRONMENTS OF ARENA AND NETFLOW

3.1 Rockwell Automation's Arena

Arena, developed by Rockwell Automation, is a simulation and automation software based on SIMAN processor and simulation language. In this simulation tool, the user places different modules and data blocks (process, decision, conditions etc.) together to build the representative model of a real business process. Connection in between these modules specifies the flow of entities. Statistical input/output data may be recorded, analyzed and reports may be generated not only with standard parameters, but also by parameters of the user's own definition.

One of the main reasons in selecting Arena as the simulation environment in this study is its integration capabilities with Microsoft technologies. It also supports open database connectivity (ODBC) which enables users to connect right into an SQL Server and retrieve data. Being Microsoft-oriented to this degree is important because the BPM tool presented in this study is designed on .NET platform. Arena includes a VBA environment in which one can edit, design and debug VB codes. Furthermore, it includes ActiveX control that provides users the ability to reference the object model library of Arena from an external application developed in Visual Basic, C++, Excel, VBA, etc. Thus Arena may be called from an external application without using Arena interface. This feature is used to develop the integration environment in the current study.

3.2 Bizitek's Netflow

Netflow is a BPM solution developed by Bizitek, a software developer in Turkey. It allows modeling related workflows according to company needs to automate workflows, follow the actions, check status reports, and see tangible results. This software consists of three main modules: Netflow Web Interface where the users log onto the system and carry out their role-specific tasks; Netflow Studio where the process owner models the process by using workflow modules and rules without doing any programming; Netflow Admin where dedicated personnel from IT department manage the whole system; role definitions, hard parameters, exception handling, database processing etc.

In Netflow Studio model, the process owner models the process and embeds the rules so that each instance follows through a pre-established logic. In doing so, however, the process owner doesn't know if the real process is mapped onto the system accurately and whether all different cases are covered. Netflow offers a "test system" where each process can be deployed first and tested against certain scenarios before live deployment takes place, but this way of testing the models may require many resources and time. Besides, testing is limited only to a number of certain scenarios which may not reflect the accurate estimations for key performance indicators.

Netflow has a built-in simulation functionality where inter-arrival times and service times (times required by organizational resources) are randomly generated from a normal distribution with a given mean and standard deviation. Obviously, the validity of the simulation models is questionable with very limited probability distributions. Lack of elaborate data collection on different attributes of entities and resources, lack of interfaces of data input and lack of flexibility as compared to other simulation and general purpose programming languages create problems when designing complex models in different areas of business.

The current study is conducted as the part of a research project on the evaluation of the BPM tool Netflow with all its features. During this study, the simulation environments of Netflow and Arena are analyzed and compared by using the evaluation framework developed by Hlupic et. al. (2007). According to the evaluation results shown in Kanalicı (2008), by the integration of Arena, the issues related to hardware and software considerations like coding aspects, software compatibility and user support will be enhanced. Furthermore, the modeling and simulation capabilities like visual aspects, efficiency, testability and statistical facilities are superior in Arena. Thus considering their simulation features and software compatibilities mentioned above, integrating Netflow and Arena seems to be a value adding approach.

4 INTEGRATION OF THE BPM AND SIMULATION TOOL

4.1 Techniques for Integrating Netflow to Arena

There are several techniques that may be used to build the interface that will transfer model data from Netflow modeling environment into Arena runtime environment. These techniques arise from the automation and integration capabilities of Arena and Netflow, as stated below:

- Developing an in-process or out-of-process DLL (or EXE).
- Direct modeling in VBA by using ActiveX automation
- Indirect modeling in VBA by using ActiveX automation (Importing Model from Database) (Retrieved from Arena help files for Arena v.10 CPR 7.0 Academic Release).

In the first technique, Arena AddOn Interface is used for developing external applications. The DLL file that is built by using this development interface can be placed on Arena's Tools menu and the simulation model is run in Arena. In this technique the users have to open Arena interface after modeling is done in Netflow and then proceed in Arena environment which is not very desirable. Hence this technique is not considered at the first place.

The second method is direct modeling in VBA by using ActiveX automation. In this technique, a code is developed within VBA workspace and is run from here. It basically retrieves data from SQL database or XML files where Netflow model data are located; and then parses this data to convert the Netflow model into Arena model by the use of ActiveX automation. Here a start button may be placed to run the VBA code so that Arena interface is not used for analyzing the simulation model.

The third technique is also implemented via an ActiveX dynamic library link (DLL). Through ActiveX, a user can write an application that converts the data stored in Netflow database to Arena model database. Then, as a second step, the program can access a built-in DLL function of Arena to

automatically import the Arena model database into Arena objects. By taking advantage of this functionality, users avoid duplicating the effort of creating a model directly in Arena.

In the second and third techniques a logic development is required to either convert the Netflow model in XML format directly to Arena objects or convert first to Arena model database and then to Arena objects. In the current study, the second technique is chosen as a starting point since it provides a direct model transfer to Arena. The indirect transfer of the third technique is still kept as an option, since there are approaches to convert a model in XML format into a standard database structure. Yet extra coding will still be necessary to convert the standard database structure into Arena database structure.

4.2 How Integration Works

The method developed to transform a model in Netflow to an executable model in Arena environment is explained in two parts: Mapping source nodes and mapping destination nodes.

4.2.1 Mapping Source Nodes

In the mapping algorithm, the code should be developed in a way that each connection should be defined by a corresponding source and a destination node with rules embedded on these connectors. Here a node represents an activity, whereas a connection represents a flow between activities. Multiple nodes originating from a single source node has to be processed in parallel. Nodes that are connected in series represent the consecutive tasks that have to be accomplished one after another. The rules on the connectors are basically the conditions that each entity is checked against and determined to advance further into the flow or disposed. In Figure 1, the XML file of a sample source node with three parallel outgoing processes is presented.

In an XML file, all the elements within < > brackets are referred to as nodes, i.e., by using a hierarchical approach, these nodes may become a parent node, a child node or both. For instance, <Steps> node is a parent node; <Condition> is a child node whereas <Direction> becomes a child node with respect to <Steps> node and a parent node with respect to <Condition> node. Every first element (word, character etc.) within brackets < > defines the corresponding tag and the rest are called attributes of that element. For example, "Type" is an attribute of the <Condition> node as well as "Script".

Before getting into details of mapping algorithm, it is necessary to identify rules in the XML program. If a rule is assigned on a connector, this is kept as "basic script" (BS) condition in the XML file. If there is no rule, then this is indicated as "All Cases" (A) in the XML file; meaning that all attributes will be passed through that connector without any check. In Figure 1, <Condition> element has two attributes: "Type", which determines whether the condition is a "Basic Script" (BS type) or "All Cases" (A type) and "Script", which holds the logical statement with the help of system variables or attributes. Here Step 1 and Step 2 are conditional, i.e., if Amount > 1000, then Step 1 is carried out and Step 2 is processed if Amount > 5000. Arena model that corresponds to this example is presented in Figure 2:

The method for converting XML data into the Arena model shown in Figure 2 is described in the following points:

- In VBA code, to transform this flow into an executable Arena model, each source node is put through a loop to extract all the destination nodes and connectors as well as condition types and condition scripts. Afterwards, source node is checked whether it is the initiator step of the flow or not; so that "Create" module is determined.
- The "Create" module is always followed by an "Assign" module, which will be used to assign values to the attributes of the entities that are extracted from "Basic Script" data. This script contains the rule operation (equation, greater than or less than) and related parameters, kept as a

string in the XML file within “Script” attribute. This string is parsed using string functions in VBA with certain logic to extract the values of the attribute in this “Assign” module and equation variables in “Decide” modules later.

```
<Steps>
<UserStep Code="Create" Start="NR" NotAssignedBehaviour="ERR" RequestComment="true"
  Help="true" Delegate="true" Iteration="1" CanSendToPool="true" StepAssignmentType="S">
<Directions>
  <Direction Code="Direction1" To="Step1" Direction="F">
    <Condition Type="BS" Script="GT(WFDATA( "MAIN", "AMOUNT" ),"1000")" />
    <From>Create</From>
  </Direction>
  <Direction Code="Direction2" To="Step2" Direction="F">
    <Condition Type="BS" Script="GT(WFDATA( "MAIN", "AMOUNT" ),"5000")" />
    <From>Create</From>
  </Direction>
  <Direction Code="Direction3" To="Step3" Direction="F">
    <Condition Type="A" />
    <From>Create</From>
  </Direction>
</Directions>
```

Figure 1. XML data kept in the SQL database corresponding to the example.

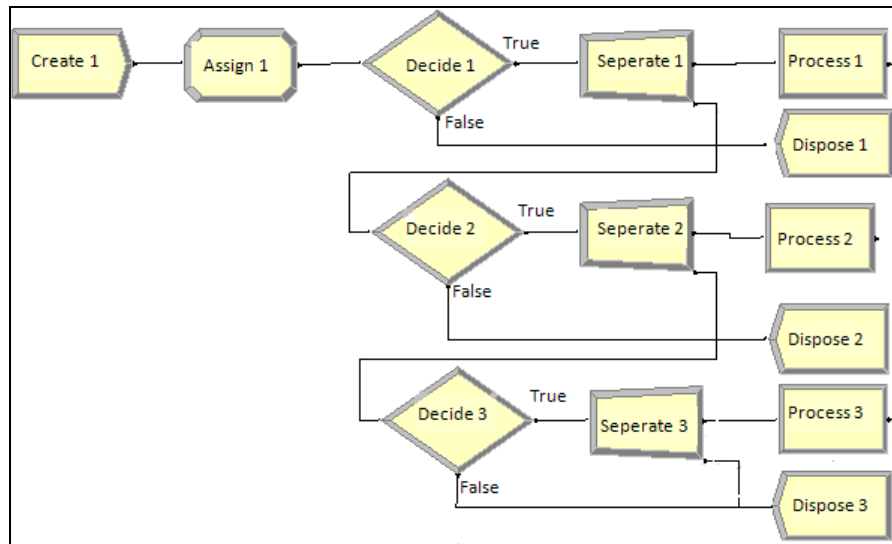


Figure 2. Model for parallel processing of steps from a single source node.

- Each connection corresponds to a group of “Decide, Separate, Process and Dispose” module group in Figure 2. This sequence of modules will serve the following functions:
 - “Decide” is configured differently for “A” type and “BS” type conditions. Basically, “A” type conditions could be seen as probabilistic conditions that are always true. However, “BS” type rules always impose that an entity must satisfy certain condition(s). In this case, decision is 2-way-conditional and the condition is supplied by parsing the string within the “Script” attributes of <Condition> element in the XML file. If the condition is satisfied, the entity will proceed to the “Separate” module where it will be duplicated for the other parallel operations and passed to the “Process” module, followed by a “Decision” module (or to the “Dispose” module if end of the flow is reached).

- “Separate” module merely acts to duplicate the incoming entity and lets it pass through the “Process” module. The duplicate would be directed to the next decision if anymore exits or to the dispose module if there is no more parallel step to be processed.
- “Process” module may be seen as the replicate of the process step in Netflow. If the connection rule is satisfied by the attribute of an entity, then that entity will proceed to “Process” module where it will be seized by a resource, processed and then released back into the flow.
- “Dispose” module is mainly used to dispose entities that do not satisfy conditions or that are loosely coupled at the end of connections.

When all the modules are created and connections are placed, they are all assigned a unique tag which contains the type of module, a counter that represents the originating step and the name of the destination step.

4.2.2 Mapping Destination Nodes

When more than one connection goes into a single destination module, then the duplicated entities should be batched up with a proper entity attribute (so that exact pairs are batched up) within a batch module. This is achieved by placing “Assign” modules after each source step to cover the existing rules on a connector going into the destination node.

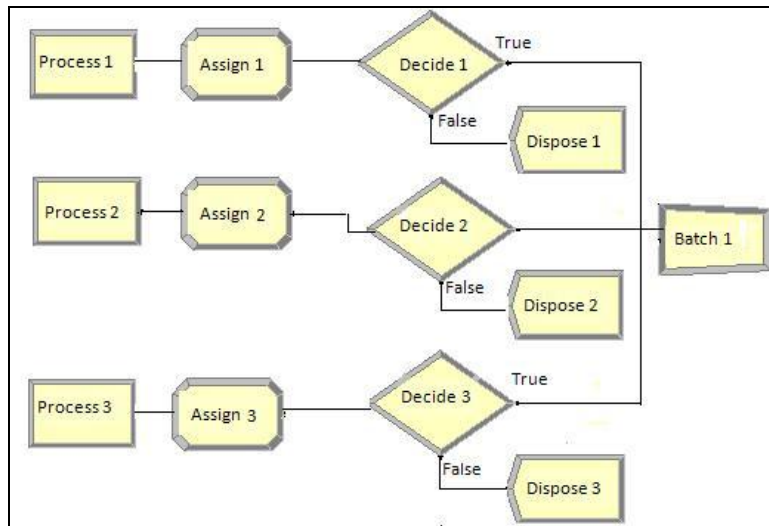


Figure 3. Arena model with multiple steps batched into one destination step.

The XML data of the model is constructed in the same structure as the single source step model. The method for converting XML data into the Arena model shown in Figure 3 is described in the following points:

- XML data is read and parsed in exactly the same way of mapping source nodes, i.e., each node is looped around all the other steps until each source and destination steps are determined by using “From” and “To” tags in the XML file.
- After each source-destination pairs are determined, an “Assign” module is assigned to each source step to identify the “BS” type conditions on these connectors. For “A” type conditions, an “Assign” module is created as well, but no assignment is made since all attributes proceed through these unconditioned connectors.
- After the attributes are assigned their values, the related conditions should be checked with a “Decide” module. Accordingly, the incoming entities are either advanced to next step or disposed.

For “A” type conditions, each entity is allowed to pass through this “Decide” module because it is valid for “All Cases”.

- “Batch” module is used to make sure that the same entity that had been duplicated for parallel processing is batched into one original entity.

4.3 Illustration and Evaluation of the Proposed Interface

In Figure 4, a typical purchasing process is shown where goods are externally procured and the requisition of this purchase needs to go through a few approval steps depending on the total amount of the required items before requisition is finally converted into a purchase order. By using the mapping algorithm and the related interface explained above, the BPM flow shown in Figure 4 is converted into a Arena simulation model in Figure 5.

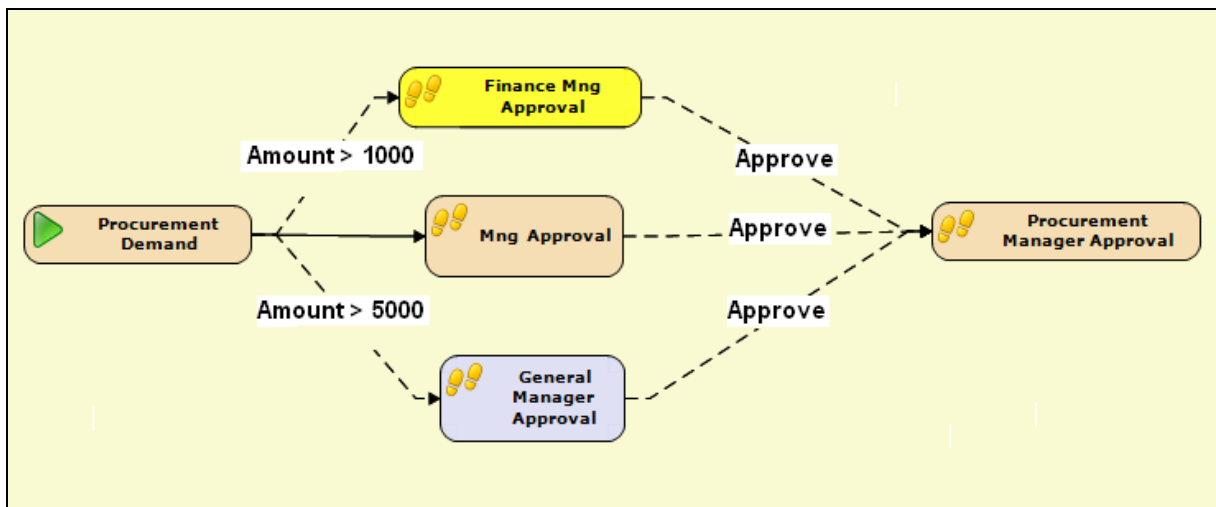


Figure 4. Sample workflow in Bizitek's Netflow.

In this example, the input distributions for the interarrival times, ordering costs, approval durations and the rate of approval for each manager are entered by the user. Formally, these data are available in the BPM database, so in application, it is possible to obtain these distributions empirically.

The generated simulation model is run in Arena to collect statistics on processes, entities and resources. The default statistical measures in Arena include the average waiting times for each activity, utilization of the resources, number of entities or tasks processed in a given period, etc. An example output for the queue statistics is shown in Figure 6.

In the next step, the generated interface is tested on different flow scenarios for verification of the integration algorithm. Scenarios are generated based on different flow structures and rules combinations. The following limitations are encountered:

- The current dialogue box and string parsing algorithm is suitable for single logical operations that include one parameter and one value (e.g. amount < 5000, amount > 1000, etc.). However, there may be many different forms of logical conditions, hence different strings, which may not be divided into logical parameters by a simple string parsing code snippet.
- A transaction may require sending an entity back to a process or decision module which had to be neglected in the current interface. Hence reverse flows should be incorporated into the conversion algorithm.

These difficulties are mainly due to the fact that all conditions are kept as strings in the XML representation of the model and this string is parsed by using string functions available in VBA. Extra coding is needed to create a user-defined class which contains the VBA version of the same methods or properties used in the underlying code of Netflow. If standard notations are fully utilized in the BPM tool under consideration then the mapping is much easier.

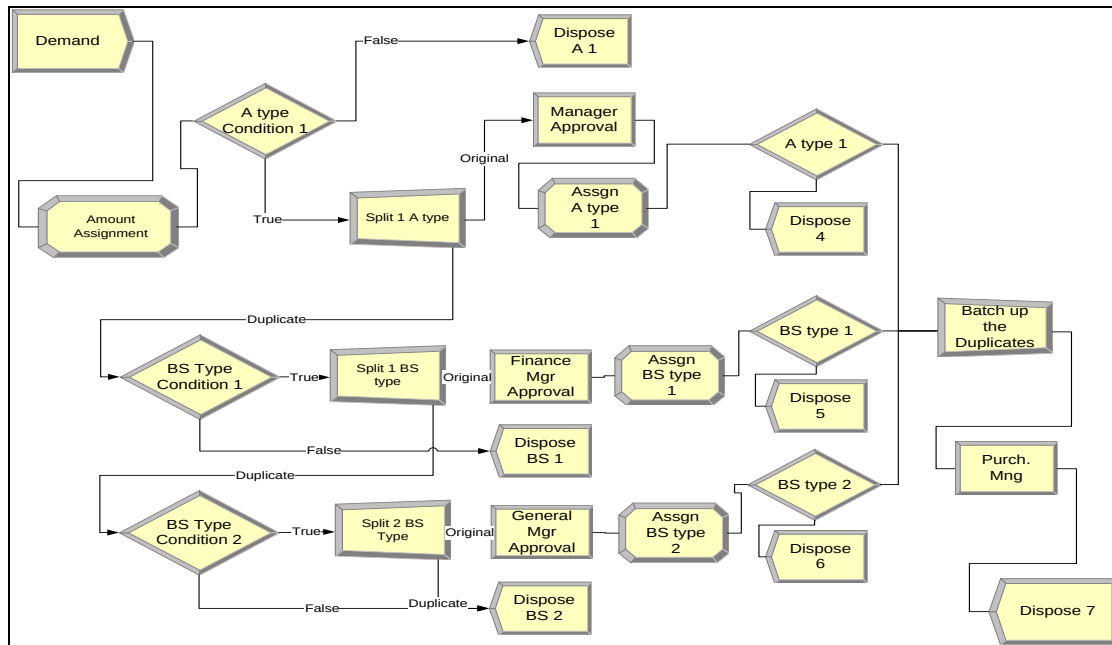


Figure 5. Sample model in Arena.

Queue						
Time						
Waiting Time	Average	Half Width	Minimum Average	Maximum Average	Minimum Value	Maximum Value
Batchup the Duplicates.Queue	6.5331	0.06	6.5125	6.5621	0.00	24.4935
Finance Mgr Approval.Queue	11.5005	0.65	11.2438	11.7696	0.00	23.5084
General Mgr Approval.Queue	0.03973995	0.17	0.00	0.1192	0.00	0.2384
Manager Approval.Queue	0.00	0.00	0.00	0.00	0.00	0.00
Purch. Mng.Queue	0.05131311	0.11	0.00	0.08385835	0.00	0.3310
Other						
Number Waiting	Average	Half Width	Minimum Average	Maximum Average	Minimum Value	Maximum Value
Batchup the Duplicates.Queue	4.6556	0.47	4.4549	4.8259	0.00	10.0000
Finance Mgr Approval.Queue	4.1604	0.27	4.0487	4.2673	0.00	9.0000
General Mgr Approval.Queue	0.00267005	0.01	0.00	0.00403614	0.00	1.0000
Manager Approval.Queue	0.00	0.00	0.00	0.00	0.00	0.00
Purch. Mng.Queue	0.00652895	0.02	0.00	0.01257875	0.00	1.0000

Figure 6: Example output: Queue statistics generated by Arena

In application of this interface for reengineering and redesign purposes, the simulation model needs to be validated as well. For the validation of the model, Netflow has to be implemented in a real system; data should be collected from the real system for the input variables like task arrival rates, durations of

activities, approval rates and output variables like average waiting times for the activities, resource utilizations, number of tasks completed, etc. In the next step, the output measures collected from the real system have to be compared with those collected from the simulated model in Arena. Obviously this requires knowledge on statistical analysis of simulation input and output.

The major benefit obtained in simulating a BPM flow in a third party simulation environment is the possibility of using this environment in its full functionality. Once the simulation model is generated, it is possible to use real input data on task arrival rates, process durations and make scenario analysis in the simulation environment. Noting that the user is not competent with the simulation software, new designs for the alternative flow scenarios should be generated in the BPM tool and then converted into the simulation environment. Furthermore, most of the simulation environments support data extraction from and into external documents such as spreadsheets, databases or text files and this ability may be used in developing output reports for the users that do not have detailed knowledge on simulation analysis.

5 CONCLUSION

It has become clearer that processes span across most of the individuals in the organizations and require efficient use of time, money and resources. Thus organizations need to have BPM tools to have better control over their complex process flows and resources. BPM Tools that are further enhanced with simulation capabilities help the organizations successfully predict the effects of new process designs in a short time and less costly way.

Simulation capabilities are integrated into BPM tools either in the form of an embedded simulation engine or as a separate module developed by themselves or through a third party simulation software. In this study, an interface is developed that can be used to map the processes designed in a BPM tool into a separate simulation software. Unfortunately, the generated algorithm is tool specific since Netflow do not utilize standard notations. It is observed that by using the XML data of the process flows and automation capabilities of VBA environment, models can be transformed into executable Arena models. The problems encountered in mapping the tasks are mainly due to the unstructured modeling environment of the current BPM tool and limitations of the VBA environment which may be recovered by extra coding.

Integration with general purpose simulation software may require high customization and get quite costly depending on the choice of the simulation software. Moreover user interfaces should be designed for providing inputs, running the model and reporting the outputs. But if this interaction is used efficiently, the benefits obtained by using a dynamic modeling environment that imitates real processes as accurate as possible are substantial.

As a future work, the current integration environment is planned to be further enhanced to allow a broader scope and content of flow models. Yet the major challenge is to work on different techniques to integrate a BPM tool into a simulation environment. For the BPM tools developed by standard notations, BPMN, applications can be easily generated to convert models stored in a database format into model database of the simulation tool. A study can be made to evaluate and compare the integration performances of this option with the current approach. Obviously such an approach requires less coding effort especially when BPMN is used.

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